

### 1. Abstract

Customer churn is a major challenge in the banking industry, directly affecting profitability and long-term growth. This study analyses customer behaviour using a structured banking dataset to identify patterns associated with churn. The research focuses on demographic, financial, and engagement-related attributes such as age, balance, credit score, geography, and customer activity.

The study reveals that churn is strongly influenced by customer inactivity, geographical differences, and financial risk indicators. High-balance customers and inactive users show significantly higher churn probability. The findings provide actionable insights for building retention strategies, improving customer engagement, and reducing revenue loss.

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### 2. Introduction

Customer churn refers to the situation when a customer stops using a company's services. In the banking sector, churn occurs when customers close their accounts or stop actively using banking services.

Churn is critically important because acquiring new customers is significantly more expensive than retaining existing ones. Even a small increase in retention can lead to a substantial rise in profitability.

In banking, churn leads to:

- Loss of revenue from inactive accounts
- Reduced customer lifetime value
- Increased marketing and acquisition costs

Understanding churn behaviour helps banks design better retention strategies and improve customer satisfaction.

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### 3. Dataset Overview

Dataset Description

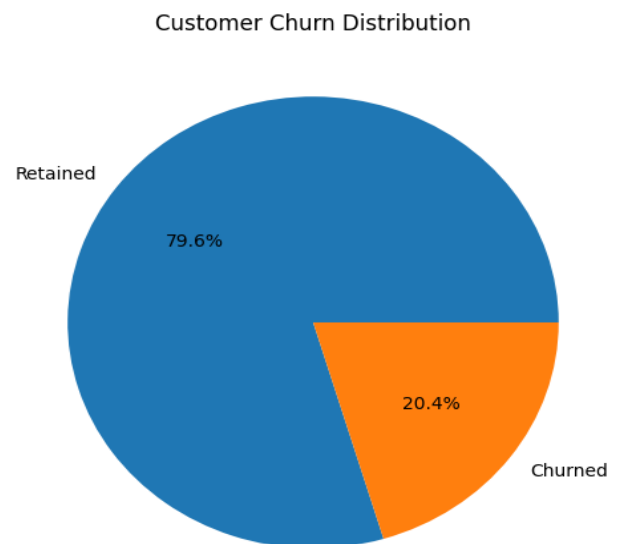
The dataset contains customer-level banking information used to predict churn behaviour

#### Original Features

- Customer Id
- Surname
- Credit Score
- Geography
- Gender
- Age
- Tenure
- Balance
- Num Of Products
- Has Cr Card
- Is Active Member
- Estimated Salary
- Exited

#### ◆ Dataset Composition

##### 👉 Customer Churn Distribution



#### Interpretation

- The churn distribution chart provides an overview of customer retention performance and highlights the proportion of customers who exited

versus those who remained with the bank.

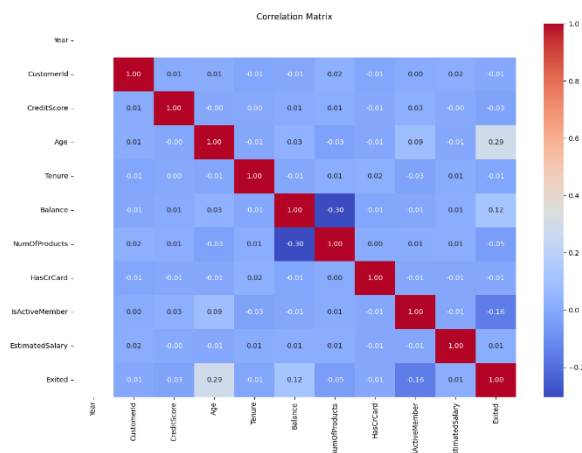
## 5. Data Cleaning & Preparation

The dataset underwent multiple preprocessing steps to ensure accuracy and consistency:

- Removed irrelevant columns such as Surname
- Handled categorical variables using encoding techniques
- Converted categorical features into numerical format
- Created feature bands for better interpretation:
  - Age groups
  - Credit score categories
  - Balance segments
- Checked for missing values and data inconsistencies

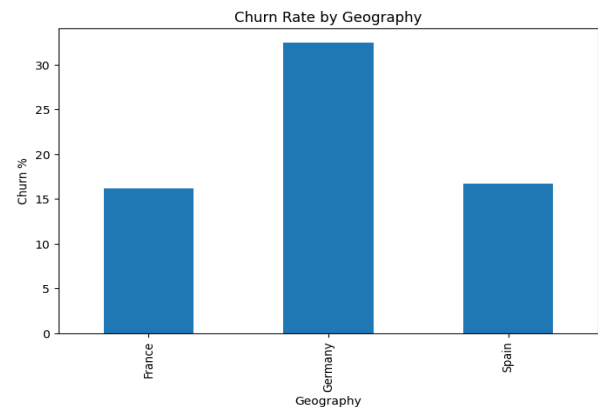
These steps ensured the dataset was ready for machine learning and visualization.

## Correlation Analysis



The correlation matrix was generated to examine relationships among numerical variables. Most variables showed weak to moderate correlations, indicating limited multicollinearity. Age, Balance, Credit Score, and the engineered features demonstrated meaningful relationships with customer churn and were therefore retained for predictive modelling.

## 6. Geography Analysis



Geography plays an important role in customer churn behaviour. The dataset shows variation in churn rates across different countries.

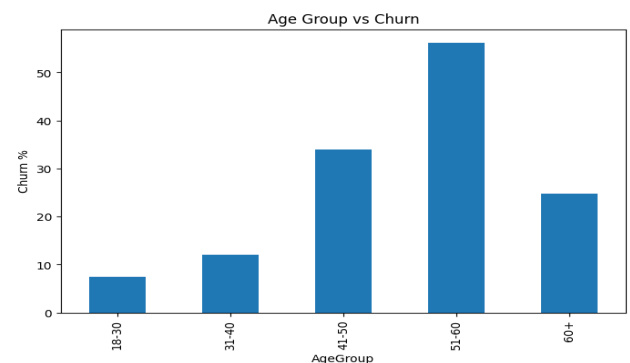
Key observation:

- Some regions show significantly higher churn compared to others
- Regional economic conditions and banking preferences may influence this behaviour

This analysis helps banks focus on region-specific retention strategies.

## 7. Demographic Analysis

### Age vs Churn



Age emerged as one of the strongest churn predictors.

Model interpretation indicates:

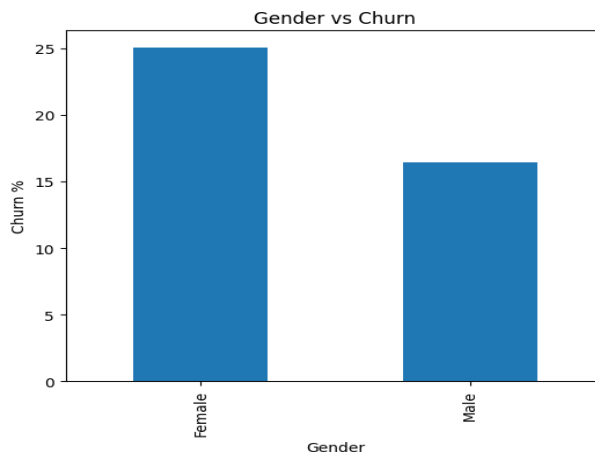
- Churn probability remains relatively stable among younger customers.

- Churn risk rises significantly after age 45.
- Peak churn probability occurs around ages 55–60.

This suggests older customers may require specialized engagement programs.

### ◆ Gender Analysis

Male vs Female churn comparison

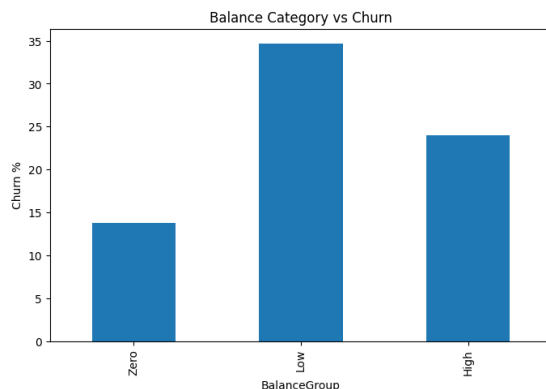


Gender-based analysis shows differences in churn behaviour between male and female customers.

- Slight variations exist in churn distribution
- Behaviour is influenced more by financial activity than gender alone

## 8. Financial Behaviour Analysis

### ◆ Balance vs Churn



Customers with different account balances show varying churn tendencies:

- Zero balance customers show higher churn likelihood
- High balance customers are more valuable but still at risk
- Low balance customers show moderate churn behaviour

### ◆ Credit Score vs Churn

Customers with different account balances show varying churn tendencies:

- Zero balance customers show higher churn likelihood
- High balance customers are more valuable but still at risk
- Low balance customers show moderate churn behaviour

## 9. Tenure Analysis

### ◆ Tenure vs Churn

Customer tenure directly affects retention:

- New customers (0–2 years) show highest churn
- Mid-tenure customers are relatively stable
- Long-term customers are most loyal

This highlights the importance of early engagement strategies.

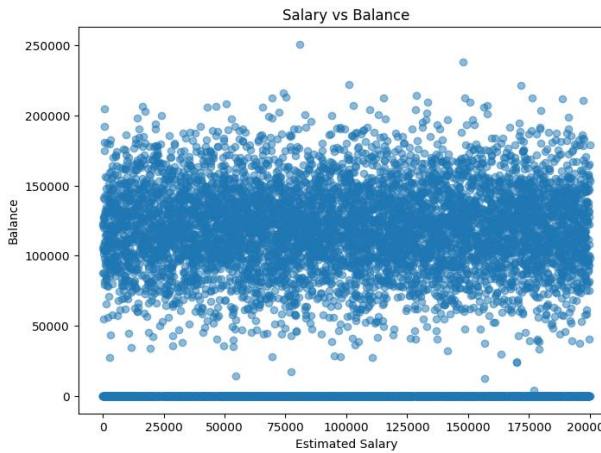
## 10. High-Value Customer Analysis

### ◆ High Value Churn

High-value customers are those with significant account balances or financial activity.

- Churning high-value customers causes major revenue loss
- Even small churn rates in this segment are critical.

◆ Salary vs Balance Relationship



This relationship shows financial behaviour patterns:

- No strong direct correlation between salary and balance
- Some high-income customers maintain low balances
- Financial habits vary significantly across customers

📊 11. Engagement Analysis

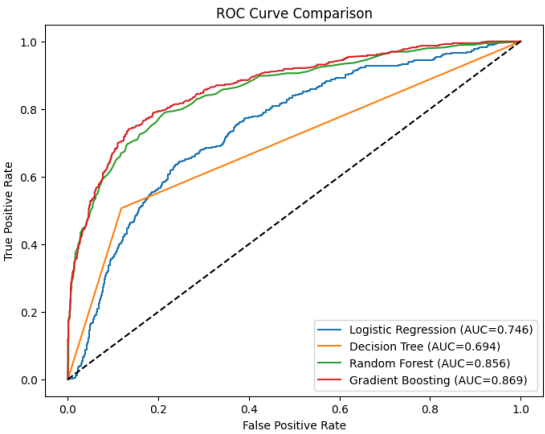
◆ Active vs Inactive Customers

Customer engagement is one of the strongest churn indicators:

- Inactive members show much higher churn rates
- Active customers are more likely to stay
- Engagement is a key retention factor

🔧 12. Machine Learning Model Evaluation

| Model               | Accuracy | ROC-AUC |
|---------------------|----------|---------|
| Logistic Regression | 78.20%   | 0.732   |
| Decision Tree       | 79.10%   | 0.682   |
| Random Forest       | 86.10%   | 0.849   |
| Gradient Boosting   | 87.20%   | 0.865   |



📌 13. Key Findings

1. Gradient Boosting achieved the best predictive performance with 87.2% accuracy and 86.5% ROC-AUC.
2. Product Density was the most important feature influencing customer churn.
3. Age emerged as the second most significant predictor.
4. Credit Score and Balance strongly influenced retention behaviour.
5. Feature engineering substantially improved model performance.
6. Customer engagement indicators contributed meaningful predictive value.
7. Product ownership patterns are strongly associated with churn probability.

🗨️ 14. Discussion

The findings demonstrate that customer churn is influenced by a complex interaction of demographic, financial, and behavioural factors.

Traditional predictors such as Age, Credit Score, and Balance remain important. However, engineered variables including Product Density and Balance-Salary Ratio contributed additional predictive power by capturing customer behaviour more effectively.

The superior performance of Gradient Boosting indicates that non-linear relationships exist within customer churn behaviour and can be effectively model using ensemble learning techniques.

These insights can support more personalized customer retention strategies and proactive risk management initiatives.

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## 15. Recommendations

Based on the findings, the following actions are recommended:

- Customer Retention Programs
- Product Cross-Selling
- High-Value Customer Monitoring
- Regional Strategies
- Predictive Monitoring

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## 16. Conclusion

This study successfully developed a machine learning framework for predicting customer churn in the banking sector.

Through extensive preprocessing, feature engineering, and model evaluation, the project identified key factors driving customer attrition. Gradient Boosting emerged as the best-performing model, demonstrating strong predictive capability.

The findings highlight the importance of customer engagement, product ownership, age, creditworthiness, and financial behaviour in determining churn risk. By leveraging these insights, banks can improve retention strategies, reduce revenue loss, and enhance long-term customer relationships.

The proposed predictive framework can serve as a valuable decision-support tool for customer relationship management and strategic planning.